Lunar image matching

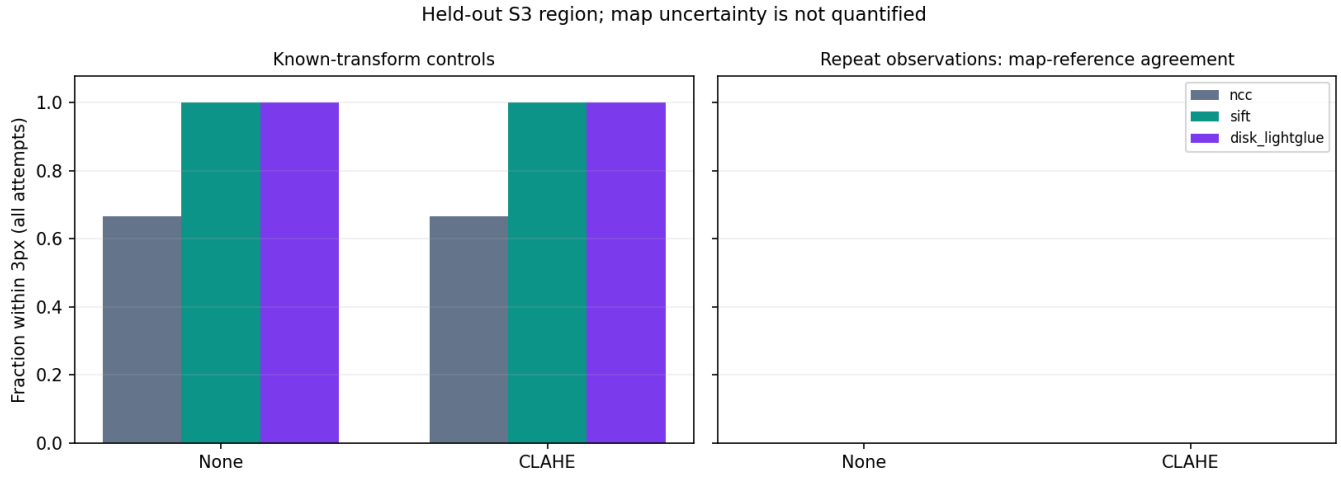
A reproducible pilot with classical and pretrained learned matching

Question. How does performance on controlled image warps compare with matching real repeat lunar observations under changing illumination?

Data. Fifteen LROC NAC map-projected crops from the published CNSFM release; 12 repeat pairs across Chang'e 3, Apollo 11 and Peak Near Shackleton. Whole scenes form reserved (S1), development (S2) and held-out (S3) splits. No model training or parameter search was performed.

Protocol. Shared 512 x 512 map grid; original nodata masks; NCC, SIFT/RANSAC and pretrained DISK/LightGlue/RANSAC; none/CLAHE ablation. Exact controls apply three known warps to one real image per scene. Failures stay in all denominators.

Reference limitation. Real observations are scored against supplied map coordinates, whose accuracy is unknown. Those numbers measure map-reference disagreement, not independent registration or navigation accuracy. Exact controls do not reproduce moving lunar shadows.

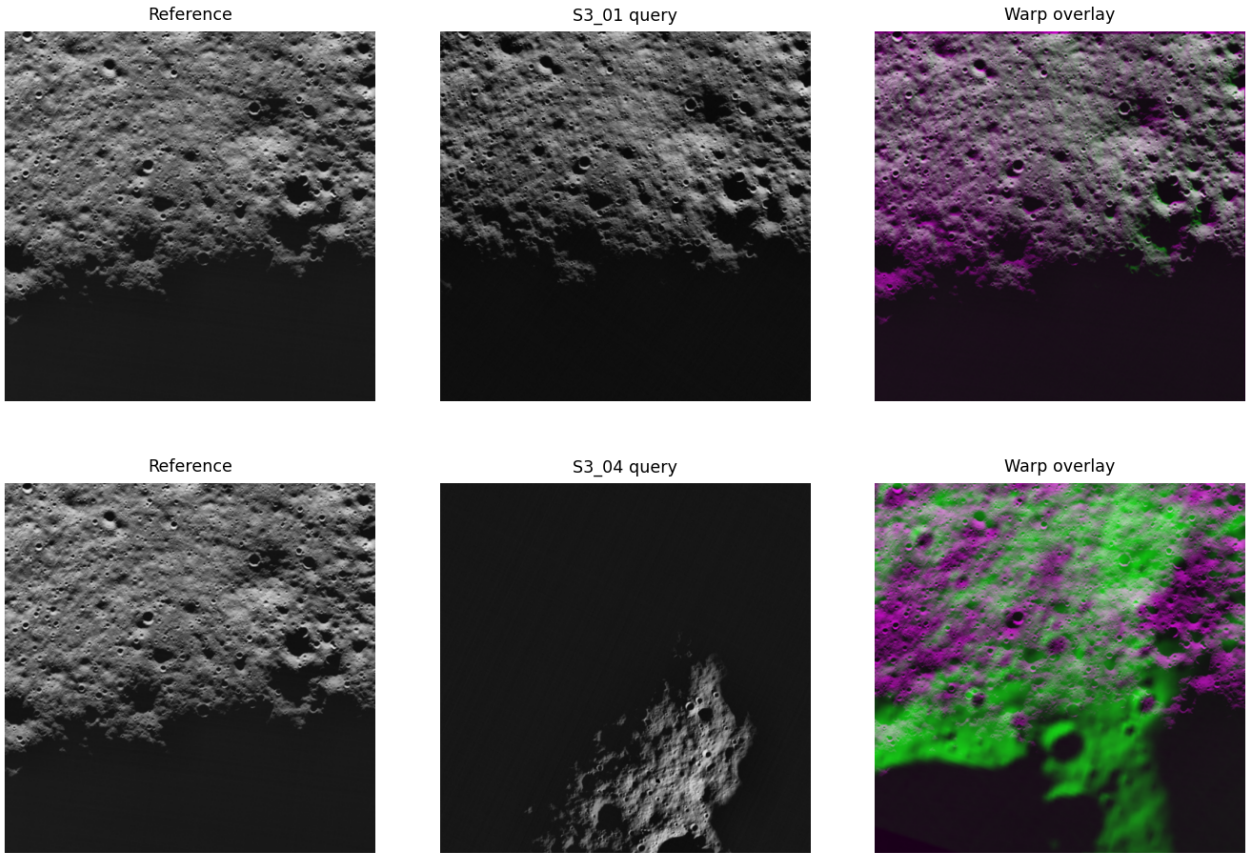


Held-out measurements

Experiment	Method / preprocessing	N	Fits	<=3px	Median px
Control	ncc / none	3	3	2	0.00
Control	ncc / clahe	3	3	2	0.00
Control	sift / none	3	3	3	0.03
Control	sift / clahe	3	3	3	0.02
Control	DISK+LG / none	3	3	3	0.05
Control	DISK+LG / clahe	3	3	3	0.06
Map ref.	ncc / none	4	2	0	4.61
Map ref.	ncc / clahe	4	2	0	4.61
Map ref.	sift / none	4	1	0	4.80
Map ref.	sift / clahe	4	1	0	4.80
Map ref.	DISK+LG / none	4	3	0	4.83
Map ref.	DISK+LG / clahe	4	3	0	4.92

Median is over scorable estimates only; the <=3px column counts every attempt. Control rotation is outside NCC's translation model. All thresholds and failure details are in results.json.

Failure analysis and reproducibility



The first and last held-out pairs by ID are shown, with DISK/LightGlue overlays. This fixed selection is qualitative; an overlay is not ground truth. Green/magenta separation shows residual disagreement.

What this study establishes. The complete pipeline runs on real lunar observations and on exact-transform controls. It records estimated transforms, failures, reference disagreement, mask support and CPU timing. Pretrained weights and source/data versions are pinned and checksummed.

Limits. Only one held-out scene and four repeat pairs; shared reference images make observations dependent. SIFT uses up to 2000 features and DISK 512. Map accuracy, solar azimuth and independent checkpoint uncertainty are unavailable. A partial-affine transform may not explain terrain relief and viewpoint changes. Timing uses one warmed repeat per condition and is descriptive.

Next scientific check. The included offline landmark-review tool supports at least ten distributed, independently annotated point pairs with picking uncertainty. Import these points and rerun the frozen evaluation before claiming navigation accuracy. No separate terrain model or LIMA reproduction is claimed.

Reproduce. Install the pinned environment; run dataset.py fetch, fetch_models.py, then evaluate.py --output runs/reproduction. Frozen configurations, asset/source hashes, per-pair results and environment metadata accompany every run.

Sources: Li et al., Remote Sensing 17(13), 2302 (2025), doi:10.3390/rs17132302; CNSFM data release. Lindenberger et al., LightGlue (ICCV 2023), arXiv:2306.13643. Tyszkiewicz et al., DISK (NeurIPS 2020), arXiv:2006.13566. See THIRD_PARTY.md for notices.